

edx Phot1x report template (2026/07)

Nicolas Casteleyn

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Abstract This report presents the results of an exercise from the **edX UBCx Phot1x: Silicon Photonics Design, Fabrication and Data Analysis** course. The work was carried out using **Ansys Interconnect** for circuit simulation, **Luceda IPKISS** for schematic capture and layout design, and **KLayout** for layout visualization and design-rule checking (DRC).

1 Introduction

Silicon photonics enables the miniaturization of free-space optical systems by integrating optical components onto a silicon chip. This approach can be effectively applied to **optical sensing applications**, which have become essential in modern engineering due to their ability to provide precise, distributed, and real-time measurements of physical parameters.

Among these **sensing techniques**, interferometric methods are particularly important. A Michelson interferometer, for example, operates by splitting an optical beam into two separate paths, reflecting the beams back, and recombining them to produce an interference pattern. This pattern contains information about variations in the optical path length caused by changes in distance, refractive index, or external perturbations such as strain and temperature.

The same interferometric principle also underpins optical frequency-domain reflectometry (OFDR), in which the analysis of interference as a function of optical frequency enables spatially resolved sensing along an optical fiber.

2 Theory

2.1 Modeling

To describe the operation of a Y-branch splitter, consider an input optical intensity I_i and an input electric-field amplitude E_i . Assuming an ideal 50:50 power splitter, the incident optical power is equally divided between the two output arms, such that

$$I_1 = I_2 = \frac{I_i}{2},$$

while the corresponding electric-field amplitudes become

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}}.$$

For the output combiner, the electric fields propagating through the two interferometer arms are recombined according to

$$E_o = \frac{E_{i_1} + E_{i_2}}{\sqrt{2}}.$$

Assuming a monochromatic plane wave

$$E = E_0 e^{i(\omega t - \beta z)},$$

where the propagation constant is

$$\beta = \frac{2\pi n}{\lambda},$$

the optical fields propagating through each arm can be expressed as follows.

$$E_{i_1} = E_1 e^{-i\beta_1 L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1} \quad (1)$$

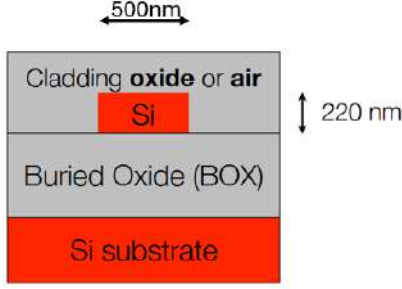


Fig. 1

Cross section model of the Si waveguide (500x220nm)

$$E_{i2} = E_2 e^{-i\beta_2 L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 L_2} \quad (2)$$

The electric field at the output of the combiner is therefore

$$E_o = \frac{E_{i1} + E_{i2}}{\sqrt{2}} = \frac{E_i}{2} (e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}) \quad (3)$$

The corresponding output optical intensity is given by

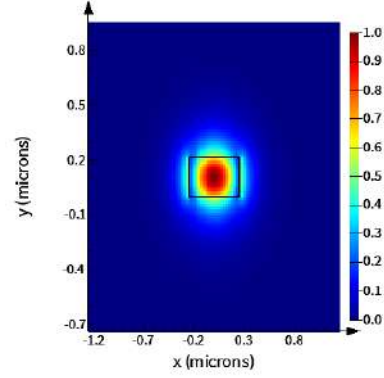
$$I_o = \frac{I_i}{4} |e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}|^2 \quad (4)$$

$$I_o = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (5)$$

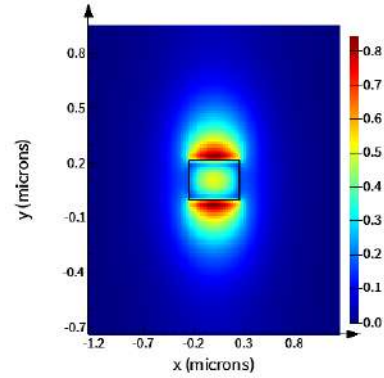
Using the SiEPIC PDK within Luceda IPKISS, a strip waveguide with a width of 500 nm and a height of 220 nm was designed, as shown in Figure 1. The fundamental guided modes supported by this waveguide were then simulated using Ansys Lumerical MODE. The resulting field distributions, corresponding to the fundamental quasi-TE and quasi-TM modes, are presented in Figures 2a and 2b, respectively.

2.1.1 Effective index and group index versus wavelength

A wavelength sweep was performed in Ansys Lumerical MODE using the silicon waveguide described previously (Figure 1). The sweep covered the wavelength range from 1.5 μm to 1.6 μm in order to evaluate the wavelength dependence of the effective index. The simulated data were exported to a CSV file and fitted with



(a)



(b)

Fig. 2: (a) quasi-TE mode (b) quasi-TM mode in Si waveguide

a second-order polynomial, providing an analytical approximation of the effective-index dispersion, given by Equation 6.

$$n_{eff}(\lambda_0) = 2.44 - 1.13 \cdot (\lambda - \lambda_0) - 0.06 \cdot (\lambda - \lambda_0)^2 \quad (6)$$

where $\lambda_0 = 1.55 \mu\text{m}$.

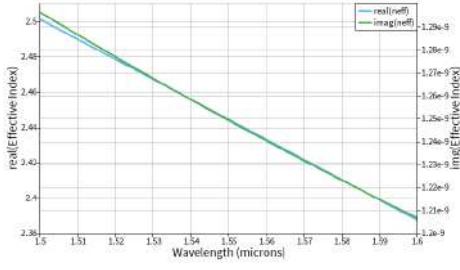
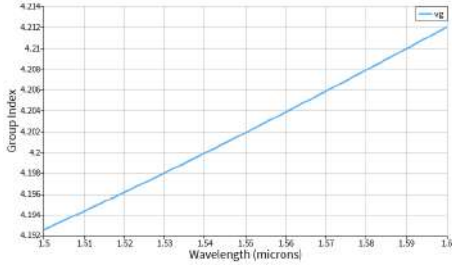
(a) Effective index n_{eff} of quasi-TE mode.(b) Group index n_g of quasi-TE mode.

Fig. 3: Quasi-TE mode characteristics as a function of wavelength: (a) effective index, and (b) group index where $n_g = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}$.

2.1.2 Insertion loss of Grating Couplers

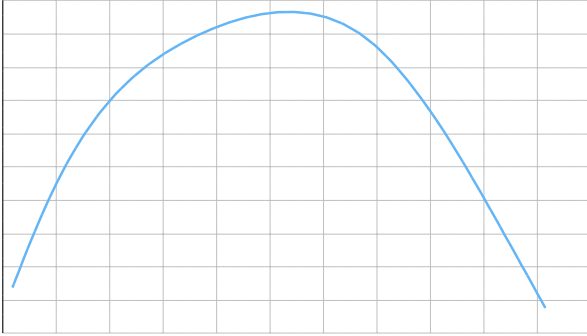


Fig. 4

Insertion loss of the grating coupler

The insertion loss of the grating couplers (Figure 4) must be taken into account, as their spectral response directly affects the measured transfer function of the interferometer. As illustrated in Figure 5, the ideal MZI spectrum (blue) is modified by the wavelength-dependent insertion losses of the input and output grating couplers, resulting in the measured spectrum shown in green.

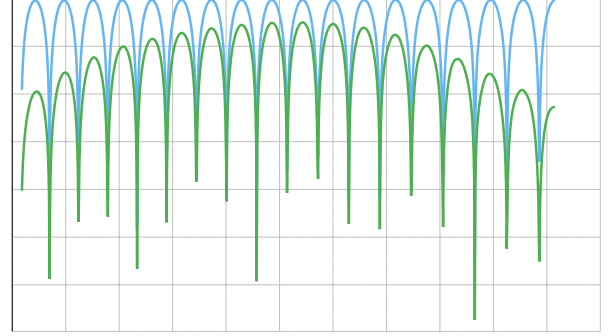


Fig. 5

MZI spectrum with (in green) or without grating couplers (in blue)

2.1.3 Mach-Zehnder interferometer (MZI)

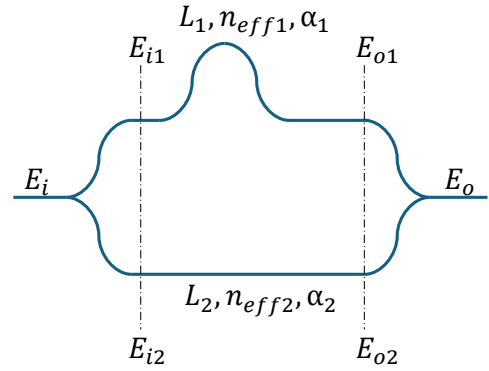


Fig. 6: Schematic of a Mach-Zehnder interferometer.

For an unbalanced Mach-Zehnder interferometer with identical waveguides, the output intensity is given by

$$I_0 = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (7)$$

In a Mach-Zehnder interferometer (MZI), the optical field is divided into two arms, propagates through each arm once, and is recombined at the output coupler. The physical path-length difference is

$$\Delta L = L_1 - L_2.$$

Since light propagates through each arm only once, the total optical path difference is simply

$$\Delta_{\text{total}} = \Delta L = L_1 - L_2.$$

FSR Formula:

$$FSR_{MZI}(\lambda) = \frac{\lambda^2}{n_g \Delta L} \quad (8)$$

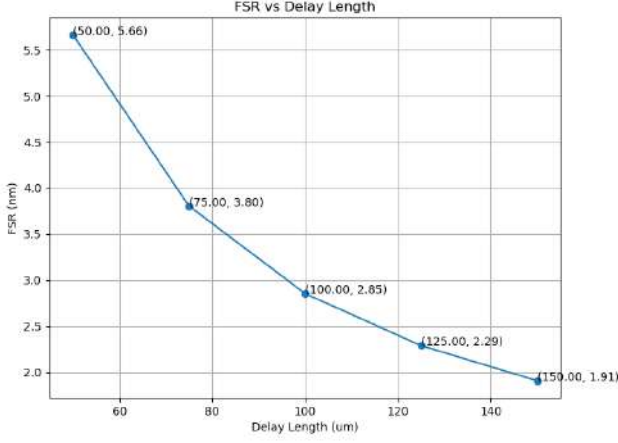


Fig. 7

FSR vs delay length for bend radius of 5μm for MZI.

2.1.4 Michelson interferometer

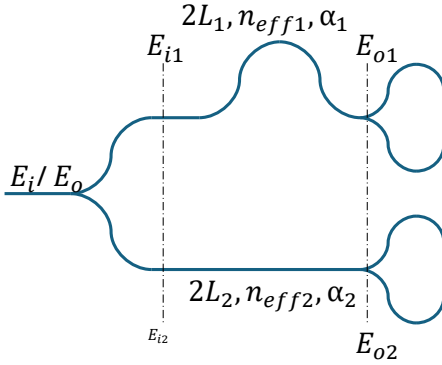


Fig. 8: Schematic of a Michelson interferometer.

In a Michelson interferometer, the optical field is divided into two arms, reflected by mirrors, and propagates back through the same waveguides before being recombined. The effective optical path difference becomes $\Delta_{\text{total}} = 2\Delta L = 2(L_1 - L_2)$. The Michelson FSR is deduced as:

$$FSR_{\text{Michelson}}(\lambda) = \frac{\lambda^2}{n_g 2\Delta L} \quad (9)$$

2.2 Simulations

2.2.1 Michelson interferometer

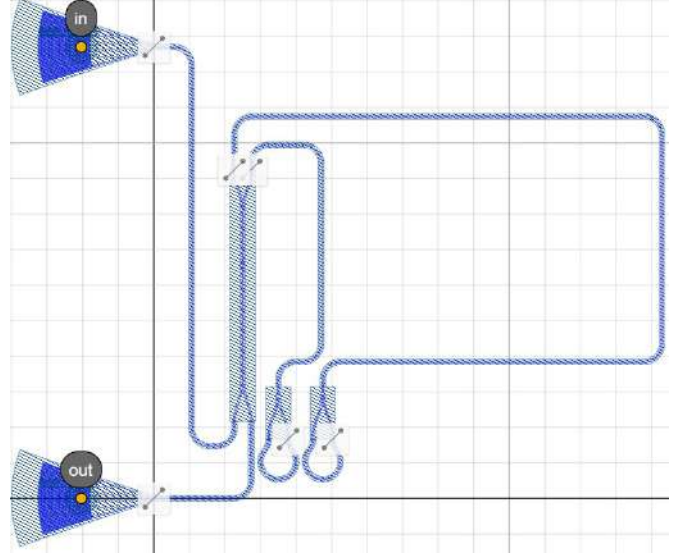


Fig. 9

Michelson interferometer in Luceda software schematic with 125 μm length delay.

2.2.2 Notes about the different layouts

- layout **A**: Michelson interferometer with a variable path-length difference and a compact directional coupler. All components operate in the TE polarization and are designed for a center wavelength of 1550 nm.

A WBG temperature sensor was also included in the layout for demonstration purposes.

- layout **B**: Michelson interferometer with a variable path-length difference and an adiabatic directional coupler. All components operate in the TE polarization and are designed for a center wavelength of 1550 nm.

- layout **C**: An attempt was made to design a polarization splitter-rotator (PSR), inspired by the Flexcompute Tidy3D example. However, the SiEPIC PDK does not provide the SiO₂ cladding required for this component. Furthermore, significant layout mismatches exist between the PSR waveguide dimensions and the standard SiEPIC waveguide cross-section, preventing a direct implementation.

Additional test structures for several photonic devices were also included in the layout.

3 Fabrication

Three chips were fabricated in this course, corresponding to the 3 different layout mentioned previously (A,B,C).

3.1 Layouts

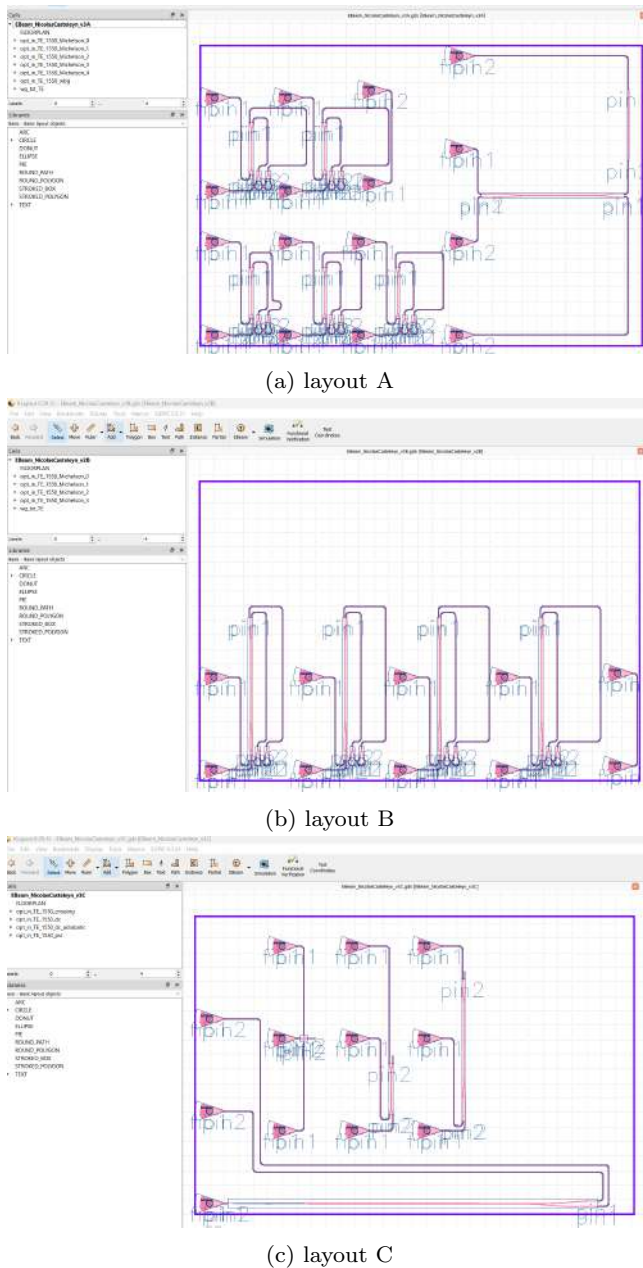


Fig. 10

3.2 Washington Nanofabrication Facility (WNF) silicon photonics process:

The devices were fabricated using 100 keV Electron Beam Lithography [2]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 μm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 $^{\circ}\text{C}$ for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 μm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 $\mu\text{C}/\text{cm}^2$ was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 $^{\circ}\text{C}$, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil.

3.3 Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a Raith EBPG 5000+ electron beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure

of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A $2.2\ \mu\text{m}$ oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C . Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

3.4 Corner Analysis

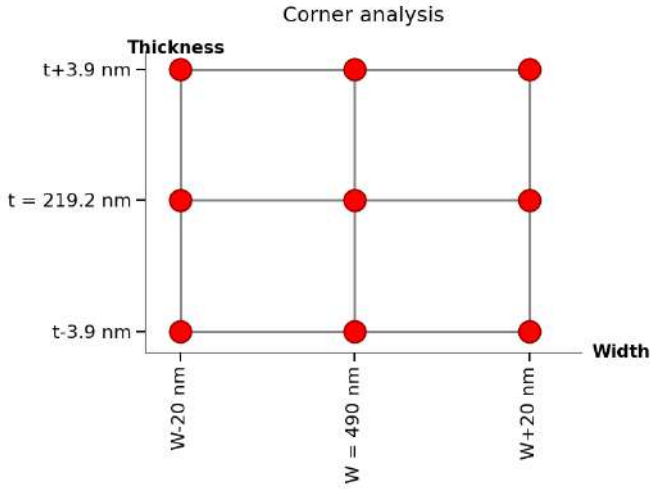
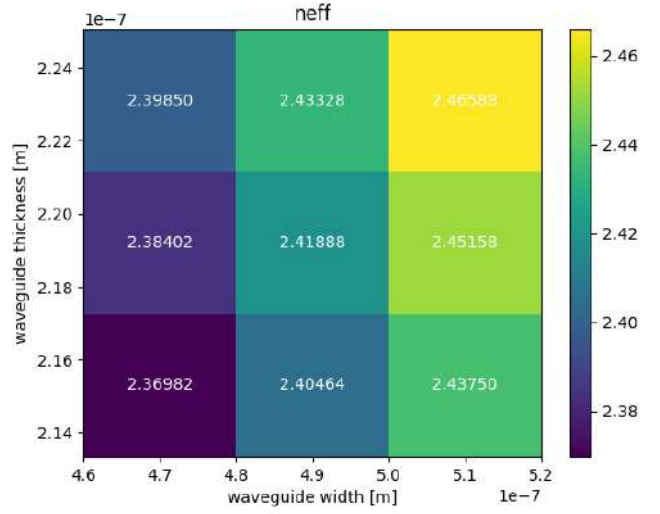
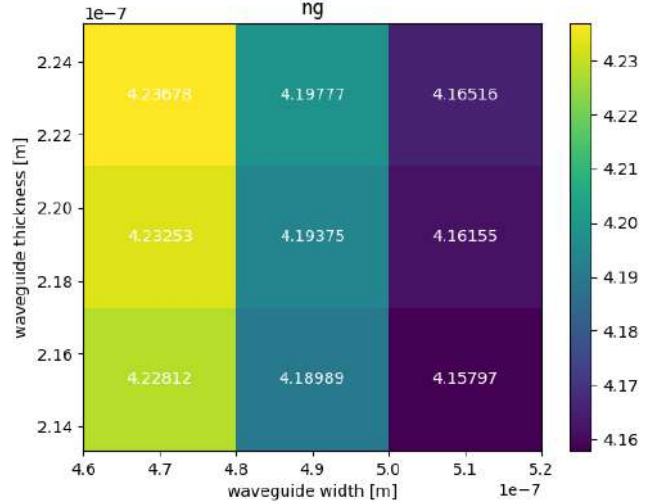


Abb. 11

Grid for corner analysis



(a) n_{eff} corner analysis



(b) n_g corner analysis

Abb. 12: Corner analysis of the waveguide parameters for (a) effective index and (b) group index.

4 Experimental Data

To characterize the devices, a custom-built automated test setup [[3]] with automated control software written in Python was used (<http://siepic.ubc.ca/probestation>). An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fiber was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [[4]]. A 90° rotation was used to inject light into the TM grating couplers [4].

A polarization maintaining fiber array was used to couple light in/out of the chip [www.plcconnections.com]

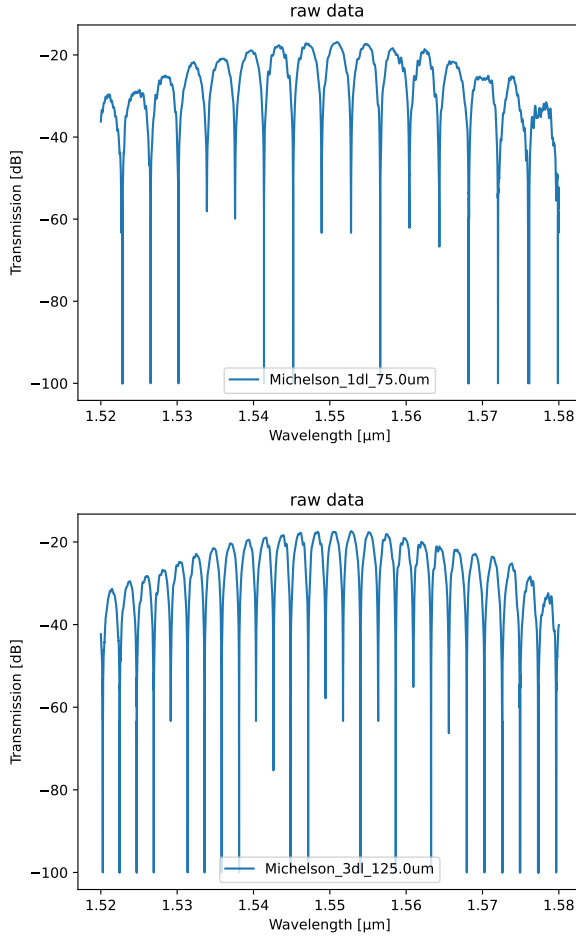


Abb. 13: Measured data on the range 1.5-1.6 μm

5 Analysis

5.1 Measure the group index

A baseline correction is first applied to the raw experimental data (Figure 13) in order to remove the spectral response of the grating couplers and normalize the interference fringes around 0 dBm. Next, a one-dimensional interpolation is performed on the corrected signal, and the wavelength vector is resampled using a uniform wavelength step. The resulting signal is then smoothed prior to peak detection, allowing an initial estimation of the free spectral range (FSR).

The initial FSR estimate is subsequently used to calculate preliminary values of the waveguide group index

and dispersion. These values serve as initial parameters for fitting the Michelson interferometer transfer-matrix model to the experimental data. Once the polynomial fitting algorithm converges, the group index and dispersion of the interferometer are extracted from the fitted transfer function. The resulting parameters are summarized in Table 1.

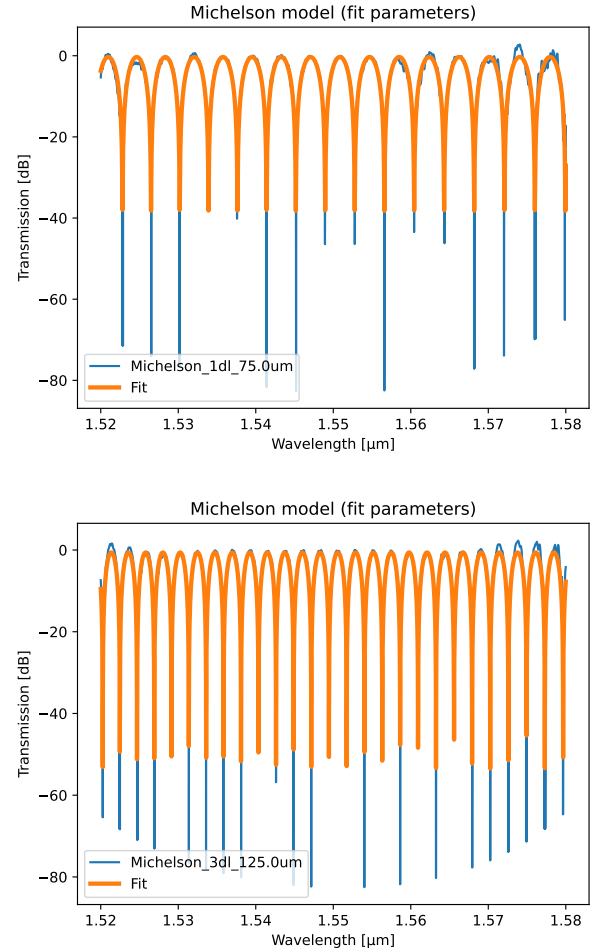


Abb. 14: Experimental transfer-function fitting used to extract the waveguide group index and dispersion.

Tabelle 1: Extracted parameters for the measured Michelson interferometers.

Michelson	Path-length difference (μm)	Group index (n_g)
1	75	4.2107
2	100	4.2030
3	125	4.2044

5.2 Comparison of experimental results with Luceda simulations

Since the interferometer layouts were designed using Luceda IPKISS, the corresponding optical simulations were also performed within the same environment. Peak detection was applied to each simulated interferometer spectrum in order to extract the theoretical FSR values. These theoretical results are then compared with the experimentally measured values, as shown in Figure 15.

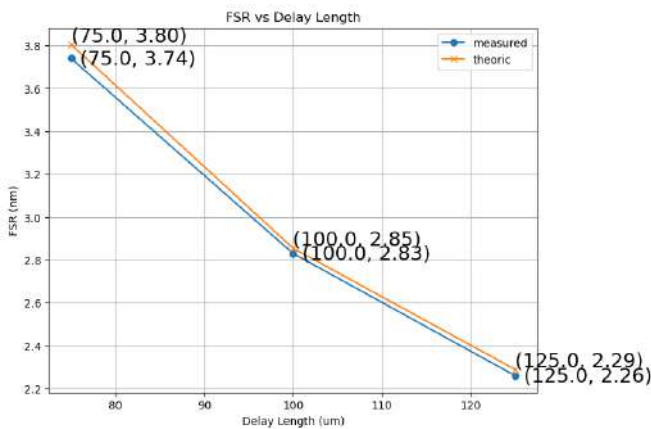


Abb. 15: Comparison of the measured and simulated free spectral ranges (FSRs) as a function of the interferometer path-length difference.

6 Conclusion

In conclusion, the measured free spectral ranges (FSRs) of the three Michelson interferometers are consistent with the results obtained from the corner analysis (Figure 12). The extracted group indices (Table 1) all fall within the predicted range,

$$4.15797 < n_g < 4.23678,$$

demonstrating good agreement between the experimental measurements and the theoretical design.

Furthermore, the extracted group indices are consistent with the initial FSR estimates obtained using the peak-detection algorithm, as illustrated in Figure 15. This agreement validates both the experimental measurements and the transfer-matrix fitting procedure used throughout the analysis.

Overall, the edX Silicon Photonics Design course provided valuable hands-on experience in the design, simulation, fabrication, and characterization of silicon photonic integrated circuits. Working with professional

tools such as Luceda IPKISS and Ansys Lumerical significantly improved my understanding of the complete photonic design workflow and reinforced the theoretical concepts presented throughout the course.

7 Acknowledgements

I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Enxiao Luan performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

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